

Multivariate Predictive Scores in the **simstudy** Pipeline: Sample Energy and Variogram Estimators

Technical Note — 0522

Abstract

This note states, with the population definitions driving the construction, exactly how the energy score and the variogram score are computed in `code/analysis/simstudy/`. Notation follows `ar2_rethink.tex` §2–§7.4; every displayed quantity is annotated with the file, function, and line range that evaluate it.

1 Setup and notation

- The holdout fixes a set of test sites $\mathcal{S}_{\text{test}} \subset \{s_1, \dots, s_{n_s}\}$, with

$$n^* := |\mathcal{S}_{\text{test}}|, \quad (1)$$

and every time $t \in \{1, \dots, T\}$ is observed at all training sites $s \notin \mathcal{S}_{\text{test}}$ (`ar2_rethink.tex` §2.3).
[`scores.R`: `obs`, l. 216; `validate`, l. 271]

- The held-out truth at time t is the vector

$$\mathbf{y}_t := (Y_t(s))_{s \in \mathcal{S}_{\text{test}}} \in \mathbb{R}^{n^*}, \quad [\mathbf{y}_t]_i = Y_t(s_i^*). \quad (2)$$

[`scores.R`: `validate <- y[!obs, ,set,setting]`; column t is $\mathbf{y}_t$]

- The forecast is the joint posterior predictive law over the test sites at time t ,

$$F_t := \text{Law}(\mathbf{Y}_t(\mathcal{S}_{\text{test}}) | \mathcal{D}), \quad \mathbf{Y}_t(\mathcal{S}_{\text{test}}) = (Y_t(s))_{s \in \mathcal{S}_{\text{test}}}, \quad (3)$$

the vector generalisation of the scalar predictive $\hat{p}(s^*, t; c)$ of `ar2_rethink.tex` eq. (predictive).

- F_t is available only through an MCMC sample: the fit stores m posterior predictive draws

$$\mathbf{x}_t^{(1)}, \dots, \mathbf{x}_t^{(m)} \overset{\text{approx.}}{\sim} F_t, \quad \mathbf{x}_t^{(j)} \in \mathbb{R}^{n^*}, \quad (4)$$

held in the array `yp` $\in \mathbb{R}^{m \times n^* \times T}$ with `[yp]j,i,t` = $x_{t,i}^{(j)}$; here $m = 10^4$. [`scores.R`: `fit$yp`, passed to `multivar_scores`, l. 296]

2 Population scoring rules

- For a forecast law F on $\mathbb{R}^{n^*}$ and an observation $\mathbf{y} \in \mathbb{R}^{n^*}$, the *energy score* [1] is

$$\text{ES}(F, \mathbf{y}) = \mathbb{E}_{X \sim F} \|X - \mathbf{y}\|_2 - \frac{1}{2} \mathbb{E}_{X, X' \overset{\text{iid}}{\sim} F} \|X - X'\|_2, \quad (5)$$

with $\|\cdot\|_2$ the Euclidean norm.

- Equation (5) is strictly proper relative to the class $\{F : \mathbb{E}_F \|X\|_2 < \infty\}$; the case $n^* = 1$ reduces it to the CRPS.
- For order $p > 0$ and nonnegative weights w_{ij} , the *variogram score* [2] is

$$\text{VS}_p(F, \mathbf{y}) = \sum_{1 \leq i < j \leq n^*} w_{ij} \left(|y_i - y_j|^p - \mathbb{E}_{X \sim F} |X_i - X_j|^p \right)^2. \quad (6)$$

- Equation (6) is proper — but *not* strictly proper — for every F with finite $2p$ -th moments: it sees F only through the pairwise p -th absolute-difference moments $\mathbb{E}_F |X_i - X_j|^p$.
- Both rules are negatively oriented (smaller is better) and act on the level Y directly.

3 Sample estimators

- Given a sample $\{\mathbf{x}^{(j)}\}_{j=1}^m$, write $\hat{F}_m := \frac{1}{m} \sum_{j=1}^m \delta_{\mathbf{x}^{(j)}}$ for its empirical measure; the estimators replace F by $\hat{F}_m$ in (5)–(6).
- The energy-score estimator (the plug-in $\text{ES}(\hat{F}_m, \mathbf{y})$) is

$$\widehat{\text{ES}}(\{\mathbf{x}^{(j)}\}_{j=1}^m, \mathbf{y}) = \frac{1}{m} \sum_{j=1}^m \|\mathbf{x}^{(j)} - \mathbf{y}\|_2 - \frac{1}{2m^2} \sum_{j=1}^m \sum_{k=1}^m \|\mathbf{x}^{(j)} - \mathbf{x}^{(k)}\|_2. \quad (7)$$

`[scoringRules::es_sample(y, dat); called in scores.R:multivar_scores, l. 242]`

- The variogram-score estimator replaces each moment $\mathbb{E}_F |X_i - X_j|^p$ by its sample mean,

$$\widehat{\text{VS}}_p(\{\mathbf{x}^{(j)}\}_{j=1}^m, \mathbf{y}) = \sum_{i=1}^{n^*} \sum_{j=1}^{n^*} w_{ij} \left(|y_i - y_j|^p - \frac{1}{m} \sum_{k=1}^m |x_i^{(k)} - x_j^{(k)}|^p \right)^2. \quad (8)$$

`[scoringRules::vs_sample(y, dat, p); called in scores.R:multivar_scores, l. 246]`

- Equation (8) sums over all *ordered* pairs; since the summand is symmetric in (i, j) and vanishes at $i = j$,

$$\widehat{\text{VS}}_p = 2 \sum_{1 \leq i < j \leq n^*} w_{ij} (\dots)^2, \quad (9)$$

i.e. twice the $\sum_{i < j}$ convention of (6); the global factor is immaterial to method ranking.

- The implementation uses uniform weights and order

$$w_{ij} \equiv 1, \quad p = 0.5, \quad (10)$$

the heavy-tail-robust choice of `ar2_rethink.tex` §7.4. `[scores.R: vs_p <- 0.5, l. 168; vs_sample called with no w argument]`

4 Predictive-draw thinning

- The double sum in (7) costs $O(m^2 n^*)$ per time point; with $m = 10^4$ stored draws this dominates runtime.

- Fix a cap M and thin the draw index to the evenly spaced set

$$\mathcal{J} = \left\{ \text{round}\left(1 + (\ell - 1) \frac{m-1}{M-1}\right) : \ell = 1, \dots, M \right\}, \quad m' := |\mathcal{J}| \leq M, \quad (11)$$

and evaluate (7)–(8) on $\{\mathbf{x}^{(j)}\}_{j \in \mathcal{J}}$; when $m \leq M$ no thinning occurs and $\mathcal{J} = \{1, \dots, m\}$.
`[scores.R: es_max_draws <- 1000L, l. 175; draw_idx, ll. 228–232]`

- Thinning leaves consistency intact: $\hat{F}_{m'} \Rightarrow F$ and the Monte Carlo error is $O(m'^{-1/2})$, negligible at $M = 10^3$ against the between-dataset spread of Section 6.

5 Per-time evaluation and the per-fit score

- At each time t the finite-truth coordinates are kept,

$$\mathcal{K}_t := \{i : [\mathbf{y}_t]_i \text{ is finite}\}, \quad (12)$$

and $\mathbf{y}_t, \mathbf{x}_t^{(j)}$ are restricted to $\mathcal{K}_t$. `[scores.R: keep <- is.finite(yt), l. 237]`

- The per-time scores are evaluated on the thinned, masked sample,

$$\widehat{\text{ES}}_t := \widehat{\text{ES}}(\{\mathbf{x}_t^{(j)}\}_{j \in \mathcal{J}}, \mathbf{y}_t), \quad \text{defined when } |\mathcal{K}_t| \geq 1, \quad (13)$$

$$\widehat{\text{VS}}_{p,t} := \widehat{\text{VS}}_p(\{\mathbf{x}_t^{(j)}\}_{j \in \mathcal{J}}, \mathbf{y}_t), \quad \text{defined when } |\mathcal{K}_t| \geq 2. \quad (14)$$

`[scores.R: multivar_scores, per-t loop, ll. 235–249]`

- Let $\mathcal{T}$ collect the times at which the score is defined and finite; the *per-fit* scores average over time,

$$\text{ES}^* = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \widehat{\text{ES}}_t, \quad \text{VS}_p^* = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \widehat{\text{VS}}_{p,t}. \quad (15)$$

`[scores.R: mean(es, na.rm=TRUE) / mean(vs, na.rm=TRUE), l. 250]`

- One fit is one (setting, method, dataset, K) tuple; the pair $(\text{ES}^*, \text{VS}_p^*)$ is stored as one cell of the score arrays. `[scores.R: energy.score[di,mi,ki] / vario.score[di,mi,ki], ll. 297–298]`

6 Cross-dataset aggregation

- Fix a method μ and basis rank K ; let $\mathcal{D}$ index the datasets with a non-missing fit. The reported summaries are

$$\overline{\text{ES}}_{\mu,K} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \text{ES}_{d,\mu,K}^*, \quad \widetilde{\text{ES}}_{\mu,K} = \text{median}_{d \in \mathcal{D}} \text{ES}_{d,\mu,K}^*, \quad (16)$$

and identically for VS_p^* . `[tables.R: mv_summary, apply(arr,c(2,3),mean/median), ll. 197–199]`

- Relative-to-Gaussian columns divide by method 1 (Gaussian) at the matched K ,

$$\text{rel_mean}_{\mu,K} = \frac{\overline{\text{ES}}_{\mu,K}}{\overline{\text{ES}}_{1,K}}, \quad \text{rel_median}_{\mu,K} = \frac{\widetilde{\text{ES}}_{\mu,K}}{\widetilde{\text{ES}}_{1,K}}, \quad (17)$$

so a value below 1 marks an improvement over the Gaussian baseline. `[tables.R: rel_of via sweep(. , 2, m["1",], "/"), ll. 200–203]`

- These four numbers per (μ, K) and per score are written to `output/tables/multivar_score<setting><suffix>.csv`. `[tables.R: multivar_table, write.csv, ll. 214–230]`

7 Properties and caveats

- The double sum of (7) retains the $j = k$ terms, so

$$\mathbb{E} \widehat{\text{ES}} = \text{ES}(F, \mathbf{y}) + \frac{1}{2m'} \mathbb{E}_F \|X - X'\|_2, \quad (18)$$

an upward bias of order $O(m'^{-1})$ that vanishes as $m' \rightarrow \infty$.

- The squared form of (8) likewise gives

$$\mathbb{E} \widehat{\text{VS}}_p = \text{VS}_p(F, \mathbf{y}) + \sum_{i,j} w_{ij} \text{Var}\left(\frac{1}{m'} \sum_k |x_i^{(k)} - x_j^{(k)}|^p\right), \quad (19)$$

an upward bias of order $O(m'^{-1})$; both estimators are consistent.

- Equations (5) and (6) score the raw level Y and are scale-sensitive: a heavy-tailed F_t inflates $\widehat{\text{ES}}_t$ without bound, so $\widehat{\text{ES}}$ may be dominated by a single dataset while $\widetilde{\text{ES}}$ stays stable — report both.
- The holdout vector (2) ranges over $\mathcal{S}_{\text{test}}$ at a *common* t , so ES^* and VS_p^* test the joint *spatial* law of F_t and are blind to cross-time dependence (`ar2_rethink.tex` §7.4: the temporal channel needs a time-block holdout).

Code map

`scores.R` Stage 1 of `code/analysis/simstudy/`.

- ll. 168, 175 — constants $p = 0.5$ (`vs_p`) and $M = 10^3$ (`es_max_draws`).
- ll. 218–251 — `multivar_scores()`: thinning (11), masking $\mathcal{K}_t$, per-time (13)–(14), time mean (15).
- ll. 296–298 — per-fit ES^* , VS_p^* written to `energy.score`, `vario.score`.

`tables.R` Stage 2 — aggregation (16).

- ll. 191–230 — `mv_summary()` builds `multivar_table`; emits `output/tables/multivar_score<setting><suffix>.csv`.

`scoringRules` CRAN package supplying the estimators `es_sample` (7) and `vs_sample` (8) [3].

References

- [1] Gneiting, T. and Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378.
- [2] Scheuerer, M. and Hamill, T. M. (2015). Variogram-based proper scoring rules for probabilistic forecasts of multivariate quantities. *Monthly Weather Review*, 143(4), 1321–1334.
- [3] Jordan, A., Krüger, F., and Lerch, S. (2019). Evaluating probabilistic forecasts with `scoringRules`. *Journal of Statistical Software*, 90(12), 1–37.